CUADebugger Pipeline

Inspect → Localize → Diagnose → Re-rollout

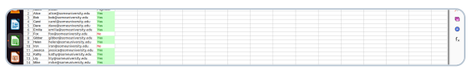
1 Failed trajectory package

TASK Add unpaid students' e-mails to the receiver field

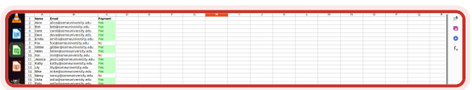
APPS Calc → Thunderbird

Input: screenshots + reasoning/action trace

- **t=5** payment record



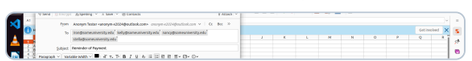
- **t=6 ROOT** invents Kelly; misses Fox



- **t=10** wrong recipients

  Iron, Kelly, Nancy, Stella; no Fox

- **t=12 FAIL** declares success; reward 0

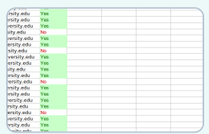


INPUT task + trajectory + terminal failure

2 Tool-guided inspection & localization

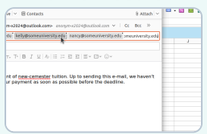
`get_step_details(6)` + `get_step_details(10)`

t=6: before



t=6 Kelly / no Fox

t=10: result



before / after | response + action | later evidence

Candidate inspection path

inspect t=6 → compare t=5/6 → inspect t=10 → localize t=6

Root: t=6 reasoning invents Kelly and omits Fox.

OUTPUT root-step candidate + grounded evidence

3 Structured RCA record

Input: candidate step + inspected evidence

ROOT STEP `root_error_step = 6` Invents Kelly; omits Fox

TAXONOMY `taxonomy_tag = P1` Visual hallucination

EVIDENCE t=5: Fox = No; t=6: Kelly/no Fox; t=10: wrong list

CORRECTION Use all No rows: Fox, Iron, Nancy, Stella

OUTPUT structured RCA record

4 Debugger-guided re-rollout

Matched continual: Claude 4.5 Sonnet; restart t=51

Continual re-rollout: Recovery / Overall

- ✓ Direct continuation: **12.50% / 61.22%**
- ✓ Self-Debug: **13.89% / 61.77%**
- ✓ Machine RCA: **21.53% / 64.82%**
- ✓ Ours + episodic memory: **25.69% / 66.48%**

Takeaway

RCA outperforms reflection; episodic memory adds **+4.16 pp recovery (+1.66 pp overall)**.

OUTPUT re-rolled trajectory + outcome

Optional Episodic Memory (+4.16 pp recovery / +1.66 pp overall)

1 · Distill failures
Trigger · failed action · correction · evidence

2 · Build query
Task · app · state · suspected action

3 · Retrieve
Top-k transferable lessons

4 · Verify
Check screenshots and trace; discard mismatches